Assignment 2 杜奕旋 12290095

1. $A=\begin{bmatrix}1&2\\1&-2\end{bmatrix}$ $B=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}$ $BA=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}\begin{bmatrix}1&2\\1&-2\end{bmatrix}=\begin{bmatrix}1&0\\0&2\end{bmatrix}$ $(BA)^9=\begin{bmatrix}1&0\\0&2^9\end{bmatrix}$

$(AB)^{10}=A(BA)^9B=\begin{bmatrix}1&2\\1&-2\end{bmatrix}\begin{bmatrix}1&0\\0&2^9\end{bmatrix}\begin{bmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}=\begin{bmatrix}1&2^{10}\\1&-2^{10}\end{bmatrix}\begin{bmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}\end{bmatrix}=\begin{bmatrix}\frac{1025}{2}&\frac{1023}{2}\\\frac{1023}{2}&\frac{1025}{2}\end{bmatrix}$

2. $a_1=\begin{bmatrix}1\\1\end{bmatrix}$ $a_2=\begin{bmatrix}1\\-1\end{bmatrix}$

$a_1a_1^T=\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1&1\end{bmatrix}$ $a_1a_2^T=\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1&-1\end{bmatrix}=\begin{bmatrix}1&-1\\1&-1\end{bmatrix}$ $a_2a_1^T=\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1&1\end{bmatrix}=\begin{bmatrix}1&1\\-1&-1\end{bmatrix}$ $a_2a_2^T=\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1&-1\end{bmatrix}=\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$

$A=k_1a_1a_1^T+k_2a_1a_2^T+k_3a_2a_1^T+k_4a_2a_2^T=k_1\begin{bmatrix}1&1\\1&1\end{bmatrix}+k_2\begin{bmatrix}1&-1\\1&-1\end{bmatrix}+k_3\begin{bmatrix}1&1\\-1&-1\end{bmatrix}+k_4\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$

$=\begin{bmatrix}k_1+k_2+k_3+k_4&k_1-k_2+k_3-k_4\\k_1+k_2-k_3-k_4&k_1-k_2-k_3+k_4\end{bmatrix}=\begin{bmatrix}-2&6\\0&4\end{bmatrix}$

$\therefore\begin{cases}k_1+k_2+k_3+k_4=-2\\k_1-k_2+k_3-k_4=6\\k_1+k_2-k_3-k_4=0\\k_1-k_2-k_3+k_4=4\end{cases}$ $\therefore\begin{cases}k_1+k_3=2\\k_1+k_2=-1\\k_1+k_4=1\\k_1-k_4=3\\k_1-k_3=2\end{cases}$ $\therefore\begin{cases}k_1=2\\k_2=-3\\k_3=0\\k_4=-1\end{cases}$

3. (a) $\because AX_0=b_0$

$\therefore EAX_0=UX_0=Eb_0=\begin{bmatrix}1&0&2&0&-1\\0&1&3&0&-2\\0&0&0&1&5\\0&0&0&0&0\end{bmatrix}\begin{bmatrix}3\\2\\0\\2\\0\end{bmatrix}=\begin{bmatrix}3\\2\\2\\0\end{bmatrix}$

$UX=Eb$ $\therefore\left[\begin{array}{ccccc|c}1&0&2&0&-1&3\\0&1&3&0&-2&2\\0&0&0&1&5&2\\0&0&0&0&0&0\end{array}\right]$

$b'=[3,2,2,0]^T$

$\therefore\begin{cases}a_1+2a_3-a_5=3\\a_2+3a_3-2a_5=2\\a_4+5a_5=2\end{cases}$ $\therefore a_4=2-5a_5$ $a_1=3-2a_3+a_5$ $a_2=2-3a_3+2a_5$

$\therefore\{(3-2\alpha+\beta,\ 2-3\alpha+2\beta,\ \alpha,\ 2-5\beta,\ \beta)\mid\alpha,\beta\text{ is real}\}$

(b) $a_1=\begin{bmatrix}2\\1\\-3\\-2\end{bmatrix}$ $a_2=\begin{bmatrix}-1\\2\\3\\1\end{bmatrix}$

$\left[\begin{array}{ccccc|c}1&0&2&0&-1&3\\0&1&3&0&-2&2\\0&0&0&1&5&2\\0&0&0&0&0&0\end{array}\right]\xrightarrow[R_3-3R_1+3R_2]{2R_1-R_2,\ 2R_2+R_1,\ R_4-2R_1+R_2}\left[\begin{array}{ccccc|c}2&-1&1&0&0&4\\1&2&8&0&-5&7\\-3&3&3&1&2&-1\\-2&1&-1&0&0&-4\end{array}\right]\xrightarrow{R_1-2R_3,\ R_2-R_3}\left[\begin{array}{ccccc|c}2&-1&1&-2&-10&0\\1&2&8&-1&-10&5\\-3&3&3&3&12&3\\2&1&-1&4&20&4\end{array}\right]$

$\therefore A=\left[\begin{array}{ccccc|c}2&-1&1&-2&-10&3\\1&2&8&-1&-10&2\\-3&3&3&3&12&0\\2&1&-1&4&20&2\end{array}\right]$

4. (a) $\operatorname{diag}(1,-2,4)\begin{bmatrix}3&4&1\\-2&0&1\\1&2&2\end{bmatrix}=\begin{bmatrix}1&0&0\\0&-2&0\\0&0&4\end{bmatrix}\begin{bmatrix}3&4&1\\-2&0&1\\1&2&2\end{bmatrix}=\begin{bmatrix}3&4&1\\4&0&-2\\4&8&8\end{bmatrix}$

(b) $\begin{bmatrix}3&4&1\\-2&0&1\\1&2&2\end{bmatrix}\operatorname{diag}(1,-2,4)=\begin{bmatrix}3&4&1\\-2&0&1\\1&2&2\end{bmatrix}\begin{bmatrix}1&0&0\\0&-2&0\\0&0&4\end{bmatrix}=\begin{bmatrix}3&-8&4\\-2&0&4\\1&-4&8\end{bmatrix}$

5. $A=\frac{1}{2}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$ $A^2=\frac{1}{4}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}=\frac{1}{4}\begin{bmatrix}2&-2\\-2&2\end{bmatrix}=\frac{1}{2}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}=A$

$\therefore A^3=A^2A=A\cdot A=A=\frac{1}{2}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$ $\therefore A^n=\frac{1}{2}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$

6. let $B=\begin{bmatrix}0&0\\1&0\end{bmatrix}$ $A=\begin{bmatrix}e&f\\g&h\end{bmatrix}$

$AB=\begin{bmatrix}f&0\\h&0\end{bmatrix}$ $BA=\begin{bmatrix}0&0\\e&f\end{bmatrix}$

when A commutes with B, $AB=BA$, $\begin{bmatrix}f&0\\h&0\end{bmatrix}=\begin{bmatrix}0&0\\e&f\end{bmatrix}$

$\therefore f=0$, $e=h$. let $e=h=a$, $g=c$

$\therefore A=\begin{bmatrix}a&0\\c&a\end{bmatrix}$

7. ① $\begin{bmatrix}I&X\\0&I\end{bmatrix}\begin{bmatrix}I&-X\\0&I\end{bmatrix}=\begin{bmatrix}I^2&0\\0&I^2\end{bmatrix}=\begin{bmatrix}I&0\\0&I\end{bmatrix}$

② $[I\ X^T][-X\ I]^T=[I\ X^T]\begin{bmatrix}-X^T\\I\end{bmatrix}=-IX^T+X^TI=0$

8. $A=\left[\begin{array}{cc|cc}2&1&3&2\\1&0&1&2\\\hline0&0&1&0\\0&0&0&1\end{array}\right]=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ $B=\left[\begin{array}{cc|c}1&-2&0\\-1&0&0\\\hline0&5&2\\-1&1&0\end{array}\right]=\begin{bmatrix}e&f\\g&h\end{bmatrix}$

$C=AB=\begin{bmatrix}ae+bg&af+bh\\ce+dg&cf+dh\end{bmatrix}=\begin{bmatrix}C_{11}&C_{12}\\C_{21}&C_{22}\end{bmatrix}$

$ae=\begin{bmatrix}1&-4\\1&-2\end{bmatrix}$ $bg=\begin{bmatrix}-2&7\\-2&7\end{bmatrix}$ $af=\begin{bmatrix}0\end{bmatrix}$ $bh=\begin{bmatrix}6\\2\end{bmatrix}$ $ce=\begin{bmatrix}0\end{bmatrix}$ $dg=\begin{bmatrix}0&5\\-1&1\end{bmatrix}$

$cf=\begin{bmatrix}0\end{bmatrix}$ $dh=\begin{bmatrix}2\\0\end{bmatrix}$

$C_{11}=\begin{bmatrix}-1&13\\-1&5\end{bmatrix}$ $C_{12}=\begin{bmatrix}6\\2\end{bmatrix}$ $C_{21}=\begin{bmatrix}0&5\\-1&1\end{bmatrix}$ $C_{22}=\begin{bmatrix}2\\0\end{bmatrix}$

$\therefore AB=\left[\begin{array}{cc|c}-1&13&6\\-1&5&2\\\hline0&5&2\\-1&1&0\end{array}\right]$

9. $(A+B)(A-B) = A^2-AB+BA-B^2$

$\Leftarrow (A-B)(A+B) = A^2+AB-BA-B^2$

$\therefore (A+B)(A-B) = (A-B)(A+B)$

$\therefore 2AB = 2BA \quad \therefore AB = BA$

$\Rightarrow \because AB = BA \quad \therefore 2AB = 2BA$

$AB-BA = BA-AB$

$A^2+AB-BA+B^2 = A^2+BA-AB+B^2$

$\therefore (A-B)(A+B) = (A+B)(A-B)$

10. $(A^T-2I)^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$ $\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 2 \end{array}\right]$

$\therefore (A^T-2I) = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} \quad \therefore A^T = 2I + \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \quad \therefore A = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

11. (a) $A^{-1}A = AA^{-1} = I \quad \therefore (A^{-1})^{-1} = A$

(b) $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I$

$(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$

12. $B = A^{-1}AB = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 1 \\ 7 & -2 & 4 \\ -1 & 2 & -1 \end{bmatrix}$

13. (a) $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \xrightarrow{R_2-aR_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & c & 1 \end{bmatrix} \xrightarrow{R_3-bR_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix} \xrightarrow{R_3-cR_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E_3E_2E_1A = U$

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & 0 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix}$

(b) $E_3E_2E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ -b+ac & -c & 1 \end{bmatrix}$

(c) $E_3E_2E_1A = U = I$

$\therefore A = E_1^{-1}E_2^{-1}E_3^{-1}I = E_1^{-1}E_2^{-1}E_3^{-1}$

$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & 0 & 1 \end{bmatrix} \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}$

$E_1^{-1}E_2^{-1}E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} = A$

14. $\begin{bmatrix} -1 & -3 & 1 \\ 1 & 4 & 1 \\ 0 & -2 & -4 \end{bmatrix} \xrightarrow[R_2+R_1]{R_1\times(-1)} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix} \xrightarrow{R_3+2R_2} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = U$

$E_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

$E_2E_1A = U \qquad L = E_1^{-1}E_2^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

$A = \begin{bmatrix} -1 & -3 & 1 \\ 1 & 4 & 1 \\ 0 & -2 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = LU$

15. $A=\begin{bmatrix}0&-2&2\\1&0&4\\-1&2&1\end{bmatrix} \xrightarrow{R_1\leftrightarrow R_2} \begin{bmatrix}1&0&4\\0&-2&2\\-1&2&1\end{bmatrix} \xrightarrow{R_3+R_1} \begin{bmatrix}1&0&4\\0&-2&2\\0&2&5\end{bmatrix} \xrightarrow{R_3+R_2} \begin{bmatrix}1&0&4\\0&-2&2\\0&0&7\end{bmatrix}$

$P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$ $PA=\begin{bmatrix}1&0&4\\0&-2&2\\-1&2&1\end{bmatrix}$ $U=\begin{bmatrix}1&0&4\\0&-2&2\\0&0&7\end{bmatrix}$

$EPA=U$ $E=\begin{bmatrix}1&0&0\\0&1&0\\1&1&1\end{bmatrix}$, $L=E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\-1&-1&1\end{bmatrix}$

$\therefore PA=\begin{bmatrix}1&0&4\\0&-2&2\\-1&2&1\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\-1&-1&1\end{bmatrix}\begin{bmatrix}1&0&4\\0&-2&2\\0&0&7\end{bmatrix}=LU$

16. $A=\begin{bmatrix}4&3&-5\\-4&-5&7\\8&6&-8\end{bmatrix}=\underbrace{\begin{bmatrix}1&0&0\\-1&1&0\\2&0&1\end{bmatrix}}_{L}\underbrace{\begin{bmatrix}4&3&-5\\0&-2&2\\0&0&2\end{bmatrix}}_{U}$

$A=LU$ $LUX=b$ $\therefore X=U^{-1}L^{-1}b$

$\left[\begin{array}{ccc|ccc}4&3&-5&1&0&0\\0&-2&2&0&1&0\\0&0&2&0&0&1\end{array}\right]\rightarrow\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&\frac{3}{8}&\frac{1}{4}\\0&1&0&0&-\frac{1}{2}&\frac{1}{2}\\0&0&1&0&0&\frac{1}{2}\end{array}\right]$ $\therefore U^{-1}=\begin{bmatrix}\frac{1}{4}&\frac{3}{8}&\frac{1}{4}\\0&-\frac{1}{2}&\frac{1}{2}\\0&0&\frac{1}{2}\end{bmatrix}$

$\left[\begin{array}{ccc|ccc}1&0&0&1&0&0\\-1&1&0&0&1&0\\2&0&1&0&0&1\end{array}\right]\rightarrow\left[\begin{array}{ccc|ccc}1&0&0&1&0&0\\0&1&0&1&1&0\\0&0&1&-2&0&1\end{array}\right]$ $\therefore L^{-1}=\begin{bmatrix}1&0&0\\1&1&0\\-2&0&1\end{bmatrix}$

$\therefore X=U^{-1}L^{-1}b=\begin{bmatrix}\frac{1}{4}&\frac{3}{8}&\frac{1}{4}\\0&-\frac{1}{2}&\frac{1}{2}\\0&0&\frac{1}{2}\end{bmatrix}\begin{bmatrix}1&0&0\\1&1&0\\-2&0&1\end{bmatrix}\begin{bmatrix}2\\-4\\6\end{bmatrix}=\begin{bmatrix}\frac{1}{8}&\frac{3}{8}&\frac{1}{4}\\-\frac{3}{2}&-\frac{1}{2}&\frac{1}{2}\\-1&0&\frac{1}{2}\end{bmatrix}\begin{bmatrix}2\\-4\\6\end{bmatrix}=\begin{bmatrix}\frac{1}{4}\\2\\1\end{bmatrix}$